Understanding the gap in evaluation frameworks and metrics for RAG-based question-answering system (Supplementary material)

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Abstract

Large Language Models (LLMs) have been extensively employed for answer generation. Particularly, Retrieval-Augmented Generation (RAG) systems have emerged as a dominant paradigm for enhancing LLM outputs with external knowledge, yet comprehensive evaluation of these systems remains fragmented and inadequate. This paper presents a systematic analysis of existing evaluation frameworks and metrics for RAG-based Question Answering (QA) systems, addressing three core research questions: the landscape of current evaluation approaches, their reliability across eight critical linguistic dimensions, and fundamental gaps requiring novel solutions. We synthesize metrics spanning classical NLP, and LLM-based evaluation paradigms, then empirically assess their robustness using a synthetically generated dataset targeting Negation, Morphology, Reordering, Paraphrasing, Active-Passive Voice, Direct-Indirect Speech, Synonyms, and Numerics. Our results demonstrate that embedding-based metrics like BERTScore exhibit superior overall performance but fail catastrophically on numerical perturbations and negation flips. We find that no single metric class suffices for comprehensive evaluation, with precision-recall asymmetries in syntactic transformations and near-chance performance of context-based metrics exposing critical vulnerabilities. These findings reveal urgent needs for hybrid evaluation frameworks, reasoning path traceability, combined pipeline assessment, and harmlessness evaluation, laying groundwork for the next generation of evaluation methodologies.

Keywords

Evaluation, LLM, RAG, Metrics, Frameworks, Question Answering (QA), LLM as Judge

1. Introduction

With our work, we aim to answer the following research questions:

- **RQ1:** What are the existing techniques and metrics used to evaluate the LLM based answer generation (in-context to a RAG based QA sytem)?
- **RQ2:** What is the reliability of these techniques in evaluating the output based on the 8 aspects of traditional NLP named Negation, Morphology, Reordering, Paraphrasing, Active-Passive voice, Direct-Indirect speech, Synonyms and Numerics? Are they enough?
- **RQ3:** What are the gaps and challenges in the existing techniques? Is there a need for new metrics or techniques?

For reproducability of the research, all the data and the analysis can be found in the GitHub link[1]: https://github.com/darkpheonix2/Metrics

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[1]https://github.com/darkpheonix2/Metrics

2. Evaluating the evaluation metrics

2.1. Fuzzy-based accuracy

The fuzzy_accuracy metric implements a sliding-window fuzzy matching algorithm designed to assess whether the ground-truth answer is semantically contained within the predicted response, accommodating minor lexical variations and word order perturbations. The algorithm first checks for exact substring containment (normalized to lowercase), returning perfect accuracy if the actual answer appears verbatim within the prediction. Failing exact match, it decomposes both strings into token sequences and slides a window of length equal to the actual answer across the predicted answer's token sequence, computing normalized similarity scores using Python's 'SequenceMatcher' (based on Ratcliff/Obershelp pattern recognition) for each window position. Formally, given predicted tokens $\mathbf{p} = [p_1, p_2, ..., p_m]$ and actual tokens $\mathbf{a} = [a_1, a_2, ..., a_n]$ where $m \geq n$, the metric computes:

$$\text{fuzzy_containment} = \max_{i \in [1, m-n+1]} \text{SequenceMatcher}\left(\mathbf{a}, \bigoplus_{j=i}^{i+n-1} p_j\right)$$

where $\oplus$ denotes token concatenation with whitespace separation. This approach addresses limitations of exact matching by rewarding partial semantic overlap, though it remains vulnerable to syntactic transformations that alter token adjacency (e.g., passive voice conversion, clause reordering) and lacks explicit semantic understanding beyond surface string similarity.

It differs fundamentally from Longest Common Subsequence (LCS) in both algorithmic approach and evaluation objective. LCS identifies the maximum-length shared subsequence between two strings, requiring only relative order preservation rather than contiguous matching: given strings X of length m and Y of length n, LCS computes $\text{LCS}(X, Y) = \max\{k : \exists i_1 < ... < i_k, j_1 < ... < j_k \text{ with } X_{i_p} = Y_{j_p}\}$ using dynamic programming with recurrence $c[i, j] = c[i-1, j-1] + 1$ if $X_i = Y_j$, else $\max(c[i-1, j], c[i, j-1])$. In contrast, fuzzy_accuracy enforces strict contiguity through its sliding window mechanism. This contiguity constraint makes fuzzy_accuracy more stringent than LCS (which permits scattered matches) but more lenient than exact matching.

2.2. Dataset construction and description

The dataset was created through a systematic synthetic data generation pipeline designed to evaluate model robustness across multiple linguistic dimensions. The process began with four established benchmark datasets: HotpotQA [2], Natural Question [3], 2WikiMultiHop[4], and Long Form QA[5]. From these sources, we initially collected 250 samples each, specifically extracting columns containing context, supporting context, and reference answers. This curated subset served as the foundation for generating synthetic variations. The expansion strategy employed an LLM (Llama-3.2-3B-Instruct[6]) to create controlled perturbations of the original answers. For each of the 250 base samples, the pipeline generated variations across eight distinct linguistic aspects plus one control condition. These aspects included semantic modifications such as enhancing semantic depth, substituting synonyms or antonyms, altering morphology, introducing negation, paraphrasing content, reordering sentence structures, converting to passive voice, transforming to indirect speech, and manipulating numerical values. This multiplicative approach-250 samples × 9 conditions (8 aspects + 1 control)-yielded a theoretical maximum of 9000 variations. Following generation, the dataset underwent manual curation to ensure quality and representation. Finally, 500 samples were selected that optimally represented all eight aspects while eliminating redundancy. Each curated sample was then manually evaluated to

[2]https://hotpotqa.github.io/
[3]https://ai.google.com/research/NaturalQuestions/dataset
[4]https://huggingface.co/datasets/xanhho/2WikiMultihopQA
[5]https://huggingface.co/datasets/akoksal/LongForm
[6]https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct

verify alignment between the original answers and their synthetic counterparts, with these assessments recorded in a dedicated "Analysis" column. Table. 1 provides an example sample of the dataset.

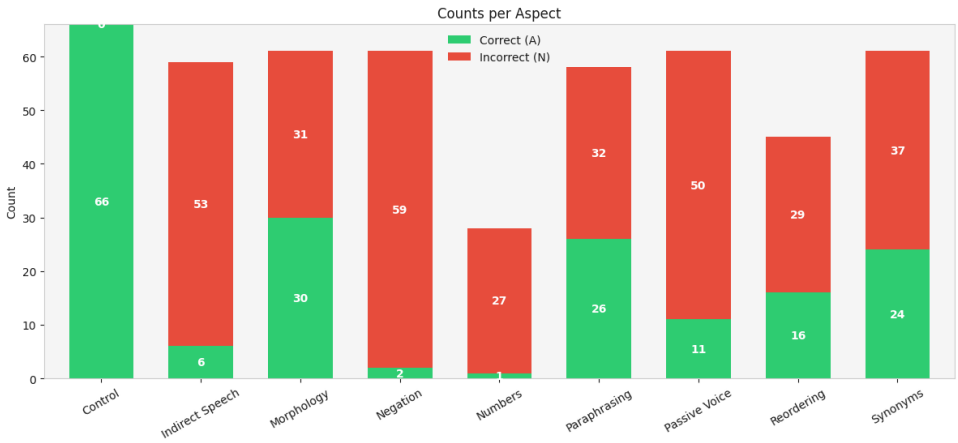


Figure 1: Dataset description

The resulting dataset structure, as shown in Fig. 1, contains multiple columns tracking the experimental design. Core fields include the original context and supporting context from the source benchmararks, the reference answer, and the generated response. Crucially, each row is tagged with an 'Aspect' column indicating which linguistic transformation was applied-ranging from specific modifications like "Negation" or "Paraphrasing" to the unaltered "Control" condition. The dataset also preserves analytical annotations from the manual evaluation phase, enabling to assess whether synthetic responses maintain semantic equivalence to their source materials despite surface-level variations. This structured format facilitates systematic analysis of how different linguistic perturbations affect model performance across question-answering tasks.

2.3. Implementation results

The aspect-specific breakdown exposes critical vulnerabilities in current evaluation methodologies. The Numbers (Fig. 2d) aspect presents the most dramatic metric failures, with Token_F1 collapsing to 0.111, Token_Precision at 0.222, and Meaning_Accuracy_Emb plummeting to 0.111 (which has highest overall AUC). This catastrophic performance reflects a fundamental challenge: numerical perturbations (changing 100 to 101) create minimal surface-level disruption while substantially altering factual content, and most metrics fail to capture this sensitivity. Conversely, Faithfulness achieves its highest score (0.854) on this aspect, suggesting that specialized fact-checking or entailment-based metrics may be necessary for numerical accuracy assessment. The near-random performance of Context_Precision and Context_Recall (0.500) across most aspects indicates these metrics provide no discriminative power for semantic equivalence evaluation, essentially performing at chance level.

For Voice (Fig. 2f), Token_Precision achieves 0.811 while Token_Recall crashes to 0.184, creating an enormous F1 disparity. This pattern emerges because passive constructions typically expand sentence length and introduce auxiliary verbs, causing precision-oriented metrics to reward the retained content words while recall metrics penalize the structural expansion. Indirect Speech (Fig. 2b) shows an even more extreme version of this phenomenon, with Token_Precision at 0.893 but Token_Recall at 0.094 (the lowest recall score across all aspects). The substantial word count increases and syntactic restructuring required for indirect speech conversion (adding "that" clauses, shifting pronouns, backshifting tenses) devastate recall-based measurements while preserving precision. BERTScore variants maintain relatively balanced performance on these aspects (0.740-0.846 range), demonstrating their superior handling of syntactic variation through semantic embedding space alignment rather than token alignment.

The Negation (Fig. 2c) exposes fundamental limitations in embedding-based semantic similarity approaches. BERTScore_R achieves only 0.297 and BERTScore_P reaches merely 0.381, among the

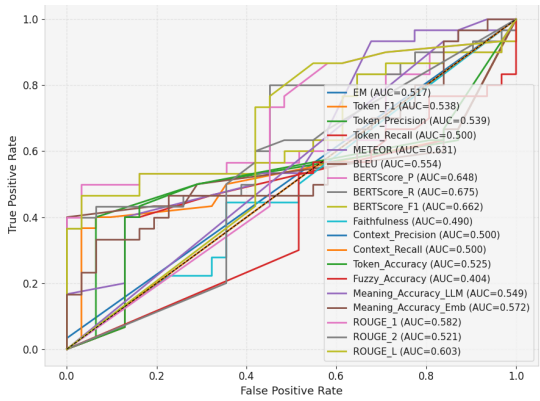
Table 1
Dataset sample for evalution of the 8 aspects

Context	Gold Answer	Model Response	Supporting context	Analysis	Aspect
(and their first in nine years) was a triumph ...	Ivan Cleary	Ivan Cleary	Ivan Cleary was ...	A	Control
(and their first in nine years) was a triumph ...	Ivan Cleary	Ivan Lahey	Ivan Cleary was ...	N	Synonyms
(and their first in nine years) was a triumph ...	Ivan Cleary	Ivan Clearys	Ivan Cleary was ...	N	Morphology
(and their first in nine years) was a triumph ...	Ivan Cleary	Ivan Cleary is never a person	Ivan Cleary was ...	N	Negation
(and their first in nine years) was a triumph ...	Ivan Cleary	Ivan Cleary is a renowned figure	Ivan Cleary was ...	N	Paraphrasing

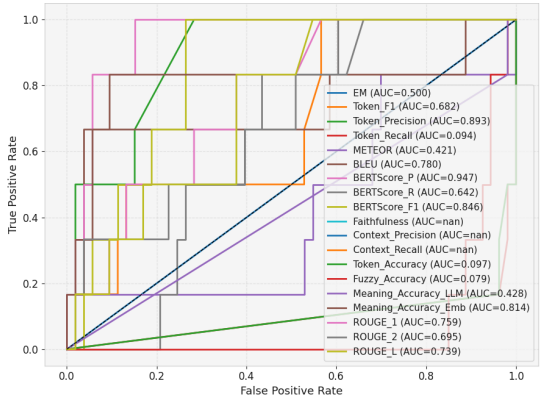
lowest scores for these metrics across any aspect. This dramatic failure occurs because negation flips the truth value of propositions while maintaining high lexical overlap and similar contextual embeddings. Interestingly, Meaning_Accuracy_LLM achieves its highest score (0.695) on this aspect, suggesting that large language model-based evaluators may possess better logical reasoning capabilities to detect polarity reversals compared to static embedding comparisons. Faithfulness also performs relatively well (0.542), indicating that entailment-based or contradiction-detection approaches may be necessary for robust negation handling.

The Synonyms (Fig. 2h) and Paraphrasing (Fig. 2e) demonstrate the most balanced metric performance, reflecting the semantic preservation goals of these transformations. Meaning_Accuracy_LLM achieves its peak performance on Synonyms (0.828), indicating that LLM-based evaluators excel when lexical substitution maintains propositional content. BERTScore_R reaches 0.771 on this aspect, substantially higher than its overall average, suggesting that recall-oriented semantic similarity is particularly effective when core concepts remain intact despite vocabulary changes. However, Fuzzy_Accuracy performs poorly on both aspects (0.331 and 0.324), demonstrating that approximate string matching fails to recognize valid paraphrases as semantically equivalent.

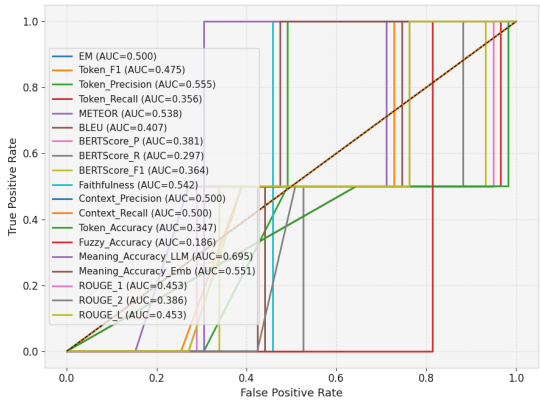
Morphology (Fig. 2a) shows relatively consistent performance across BERTScore variants (0.648-0.675) and traditional metrics, with METEOR achieving 0.631 through its synonym-aware matching. Reordering (Fig. 2g) demonstrates the critical importance of handling non-sequential similarity, with Token_Recall at a mere 0.406 while BERTScore maintains more robust performance (0.416-0.444). The



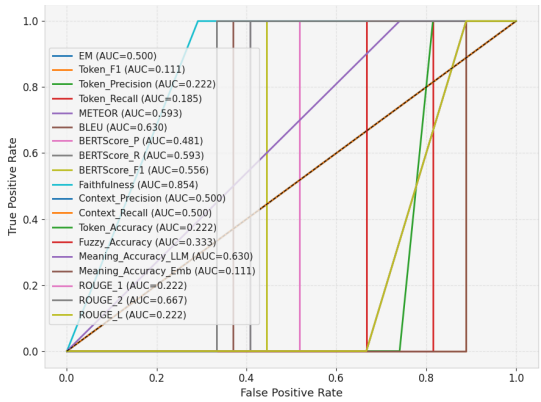
(a) Morphology



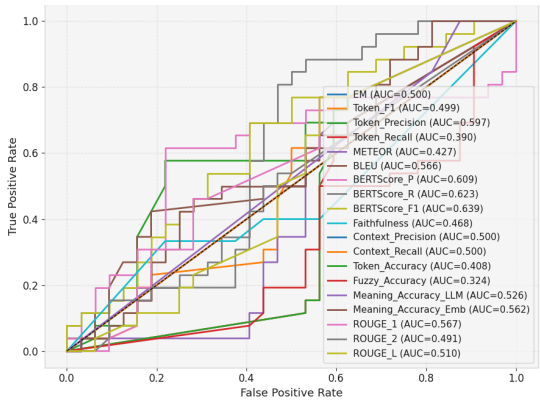
(b) Direct-Indirect Speech



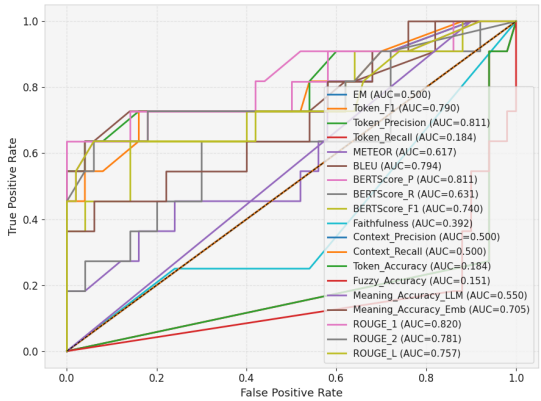
(c) Negation



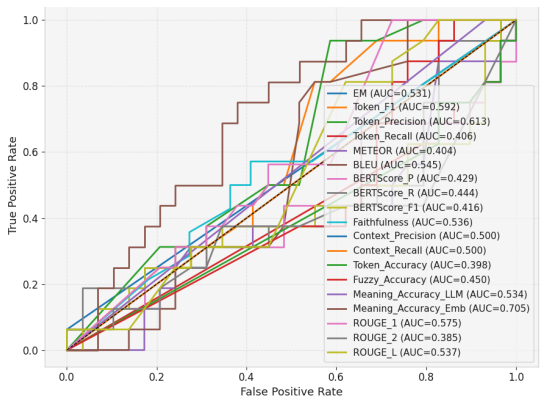
(d) Numerics



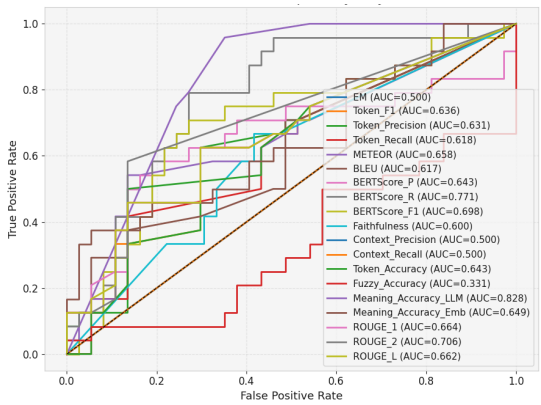
(e) Paraphrasing



(f) Active-Passive voice



(g) Reordering



(h) Synonyms

Figure 2: Aspect wise AUC-ROC analysis

Meaning_Accuracy_Emb score of 0.705 on Reordering suggests that sentence-level embedding averaging may partially compensate for word order changes, though this remains below its overall average.

These results collectively argue for a hybrid evaluation framework that combines complementary metric classes. The consistent superiority of BERTScore variants suggests that contextualized embedding similarity should form the foundation of semantic equivalence assessment. However, the specific failures (numerical sensitivity, negation detection, and extreme syntactic transformations) indicate that specialized modules are necessary. Fact-checking components for numerical accuracy, natural language inference models for negation and contradiction detection, and potentially LLM-based judges for complex semantic reasoning would address the blind spots of embedding-based approaches. The complete inadequacy of the non-discriminative performance of context-based metrics (Context_Precision, Context_Recall) suggests these should be excluded from robust evaluation protocols. The dramatic precision-recall trade-offs observed in syntactic transformations also imply that F1 aggregation may obscure important metric behaviors, and that aspect-specific metric selection may ultimately prove more effective than universal metric application.

The Spearman correlation analysis (Fig. 3) reveals a striking hierarchy of metric reliability when assessed against human judgments of semantic equivalence, with exact match (EM) unexpectedly dominating despite its notorious brittleness, suggesting that human annotators in this study privileged surface-level correctness over semantic flexibility. BERTScore variants cluster tightly in the $r_s = 0.367$–0.410 range, demonstrating that embedding-based similarity captures moderate but incomplete alignment with human perception, while the divergence between BERTScore_P ($r_s = 0.4102$) and BERTScore_R ($r_s = 0.3671$) indicates that precision oriented semantic matching resonates more strongly with human judgment than recall-oriented coverage. The precipitous decline of n-gram metrics ROUGE_2 and BLEU confirms that strict bigram matching poorly approximates human semantic assessment, while METEOR's weak performance despite its synonym-awareness suggests that its heuristic weighting scheme fails to compensate for underlying n-gram limitations. Most critically, Faithfulness exhibits negligible correlation ($r_s = 0.0175, p = 0.7230$), indicating that its fact verification signal operates orthogonally to human semantic equivalence judgments.

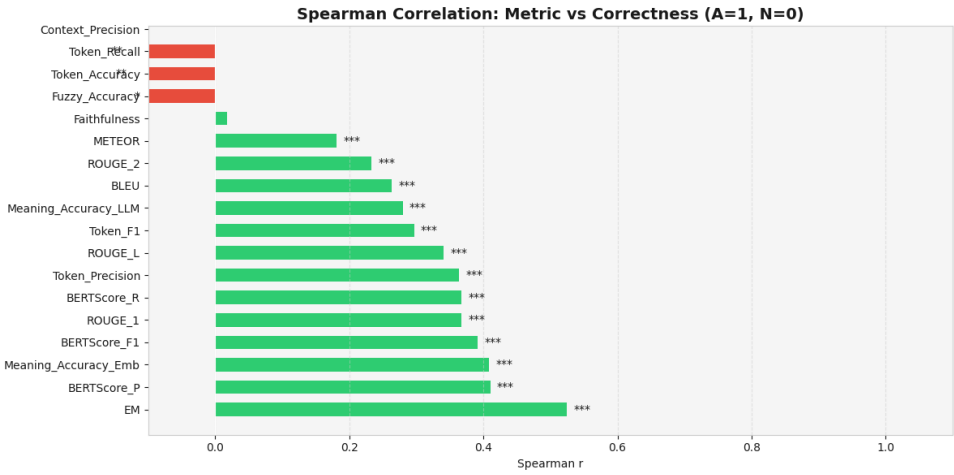


Figure 3: Spearman correlation (Higher the better)

The aspect sensitivity heatmap (Fig. 4) and false-high rate analysis (Fig. 5) collectively expose critical failure modes in RAG evaluation metrics, revealing how embedding-based approaches generate misleadingly optimistic scores when semantic content has been fundamentally altered. The heatmap visualizes normalized sensitivity scores across eight linguistic transformation aspects, with darker cells indicating higher metric responsiveness to specific perturbations. Strikingly, the Numbers aspect shows extreme sensitivity to Faithfulness while Meaning_Accuracy_Emb collapses to near-zero, confirming that embedding-based metrics fail to detect numerical perturbations that preserve surface similarity. Conversely, Synonyms demonstrates the highest sensitivity to Meaning_Accuracy_LLM (0.43), indicat-

ing that LLM-based evaluators excel at recognizing lexical substitution while maintaining propositional content, yet this same metric shows minimal responsiveness to Passive Voice (0.18) and Negation (0.13), exposing its blind spot to syntactic transformations that alter sentence structure without changing lexical items.

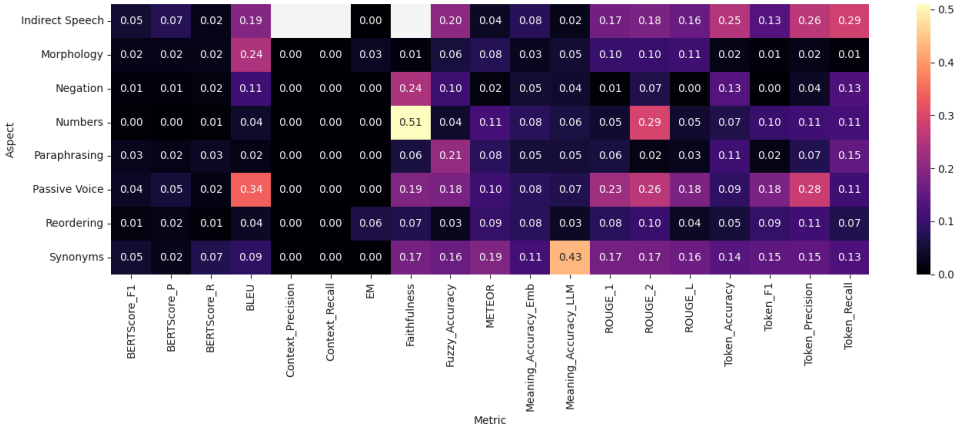


Figure 4: Aspect Senstivity Value

The false-high rate table quantifies this danger more explicitly, measuring the percentage of meaning-altering transformations (negation flips, number changes) that receive incorrectly high scores (normalized score > 0.8). BERTScore variants cluster at 95-97% false-high rates, revealing that their strong overall AUC performance masks a dangerous tendency to assign near-perfect scores to semantically corrupted outputs—precisely the failure mode most damaging to reliable evaluation.

The middle tier presents equally concerning patterns: Fuzzy_Accuracy (72.33%), Meaning_Accuracy_LLM (71.07%), and token-level metrics (67.30%) demonstrate substantial false-positive risk, indicating that answer containment strategies and LLM-based judges frequently misclassify negated or numerically altered content as semantically equivalent. Only Faithfulness (30.85%) and traditional n-gram metrics (ROUGE_1 at 16.98%, ROUGE_L at 14.47%) maintain moderate discriminative power, with EM achieving perfect specificity (0.00% false-high rate) through its binary exact matching, though this comes at the cost of rejecting all valid paraphrases. The BLEU score's anomalously low false-high rate (2.20%) reflects its geometric mean formulation penalizing any length deviation or n-gram mismatch so severely that even minor transformations trigger score collapse, making it overly conservative rather than genuinely discerning.

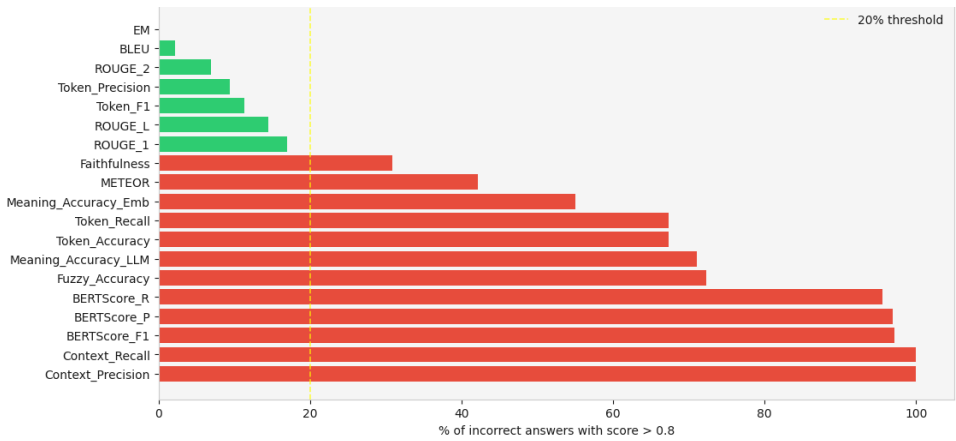


Figure 5:

These findings fundamentally challenge current evaluation practices: metrics achieving high AUC

through broad dynamic range (BERTScore, embedding-based approaches) may be precisely those most dangerous for deployment, as their high scores on corrupted outputs provide false assurance of quality. The inverse relationship between overall AUC and false-high rates where top performing metrics by discrimination exhibit worst specificity suggests that evaluation reliability requires balancing sensitivity to semantic preservation with robustness to adversarial perturbations. For RAG systems in high-stakes domains (financial reporting, medical question answering), the 95%+ false-high rates of embedding-based metrics represent an unacceptable risk, necessitating hybrid architectures that combine BERTScore's general robustness with specialized fact-verification modules (Faithfulness) and explicit contradiction detection for negation and numerical accuracy.

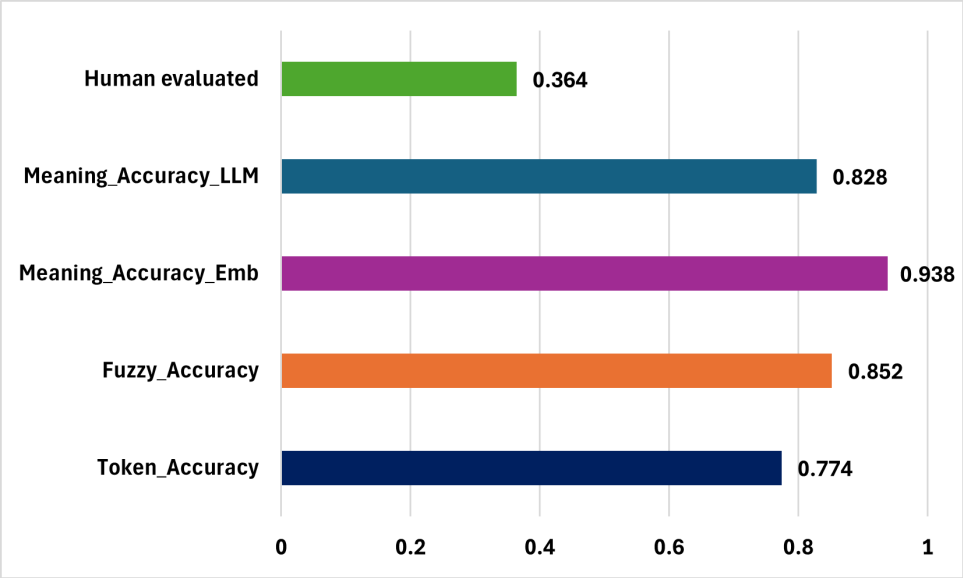


Figure 6: Comparison of accuracy by various methods

Fig. 6 reveals a critical diagnostic of metric calibration by comparing automated scoring behavior against human judgment distributions. The analysis computes the proportion of samples where each metric assigns a score above 0.5 (treating this as a binary "pass" threshold) and contrasts these with the actual human acceptance rate (0.364 or 36.4%).

The results expose severe over-acceptance bias across all automated metrics: Token_Accuracy passes 77.4% of samples, Fuzzy_Accuracy 85.2%, Meaning_Accuracy_LLM 82.8%, and Meaning_Accuracy_Emb an extreme 93.8% all dramatically exceeding the human acceptance rate of 36.4%. This discrepancy indicates that automated metrics operate with systematically inflated score distributions, where their 0.5 threshold functions as a lenient barrier that admits the majority of outputs regardless of actual semantic quality.

The calibration failure is most pronounced for Meaning_Accuracy_Emb (93.8% pass rate vs. 36.4% human rate), suggesting that embedding-based similarity measures assign moderate-to-high scores even for substantially degraded or transformed outputs, lacking the discriminative sharpness to separate acceptable from unacceptable generations. Conversely, Token_Accuracy, despite its brittleness, shows the closest alignment to human judgment (77.4% vs. 36.4%), though still severely over-permissive.

This pattern implies that threshold-based metric deployment in production RAG systems risks silent quality degradation automated evaluations may report high pass rates while human reviewers would reject nearly two-thirds of outputs. The 57.4 percentage point gap between the most calibrated automated metric and human judgment (93.8% - 36.4% = 57.4%) underscores the necessity of metric recalibration, threshold adjustment, or human-in-the-loop validation for reliable quality assurance.

3. Discussion

The empirical findings from this study fundamentally challenge the prevailing assumption that single-metric evaluation suffices for assessing RAG system outputs, revealing instead a landscape of complementary weaknesses where each metric class excels in specific linguistic dimensions while failing catastrophically in others. The AUC analysis demonstrates that no existing metric achieves acceptable discrimination (AUC > 0.5) across all eight traditional NLP aspects with even the strongest performers exhibiting critical blind spots that render them unsuitable for standalone deployment. Specifically, Token_Precision, BLEU, Meaning_Accuracy_LLM, and Meaning_Accuracy_Emb each fail on only one aspect, suggesting that strategic combination rather than metric selection offers the most viable path toward comprehensive evaluation. The harmonic mean proposal for Meaning_Accuracy_LLM and Meaning_Accuracy_Emb exemplifies this compensatory logic: where Meaning_Accuracy_LLM underperforms on Direct-Indirect Speech transformations due to its sensitivity to syntactic restructuring, Meaning_Accuracy_Emb maintains robustness through embedding space alignment, while conversely, Meaning_Accuracy_Emb's catastrophic failure on Numerics (AUC = 0.111) is offset by Meaning_Accuracy_LLM's moderate performance (AUC = 0.630). This architectural insight extends to Faithfulness, which demonstrates unique strength in numerical fact verification (AUC = 0.854) but weak general semantic discrimination (AUC = 0.510), indicating its optimal deployment as a specialized module for grounding verification rather than a standalone quality metric.

The calibration analysis exposes a second critical gap: systematic over-acceptance bias across all automated metrics, with pass rates at the 0.5 threshold ranging from 77.4% (Token_Accuracy) to 93.8% (Meaning_Accuracy_Emb) against a human acceptance rate of merely 36.4%. This 40-60 percentage point divergence reveals that metric score distributions are poorly aligned with human judgment thresholds, risking silent quality degradation in production systems where automated evaluation reports high pass rates while human reviewers would reject the majority of outputs. The false-high rate analysis compounds this concern, showing that BERTScore variants assign scores > 0.8 to 95-97% of meaning-altering transformations, providing false assurance of quality precisely when evaluation is most critical. These calibration failures suggest that current metrics lack the sharpness necessary for reliable binary quality decisions, operating instead as coarse similarity measures that conflate surface lexical overlap with semantic fidelity.

3.1. Gaps

Several structural gaps in current evaluation paradigms emerge from this analysis. First, the complete absence of indexing quality assessment represents a fundamental oversight current frameworks treat retrieval and generation as discrete black boxes, ignoring that retrieval errors propagate irreversibly through generation regardless of downstream metric performance. Second, the aspect sensitivity heatmap reveals that no metric adequately addresses compositional reasoning or multi-hop inference, dimensions critical for complex question answering where answers require synthesizing information across multiple retrieved passages. Third, the negative Spearman correlations between token-level matching metrics and human judgment (-0.102 to -0.128) indicate that metrics rewarding lexical overlap actively anti-correlate with human perception when evaluating linguistically diverse but semantically preserved outputs, suggesting that current optimization objectives may inadvertently penalize desirable linguistic variation.

3.2. Directions

Future evaluation frameworks must address these limitations through several architectural innovations.

Dynamic metric selection based on query classification could route evaluation to aspect-specific metrics-deploying Faithfulness for numerical queries, Meaning_Accuracy_LLM for synonym-heavy domains, and ensemble methods for syntactically complex transformations.

Calibration-aware thresholds should replace fixed 0.5 cutoffs with aspect-specific operating points

derived from validation set distributions, ensuring that reported pass rates align with actual quality levels.

Most critically, **reasoning trace evaluation** must supplement answer assessment, incorporating retrieval path verification, coherence checking across multi-hop chains, and step-wise faithfulness metrics that validate whether generated responses accurately reflect the retrieved context rather than hallucinating connections. The incorporation of confidence intervals and Likert-scale comparative evaluation, as suggested in recent work, could further enhance reliability by leveraging LLMs' superior performance in relative versus absolute judgment tasks.

4. Conclusion

This study presents a comprehensive analysis of evaluation metrics for Retrieval-Augmented Generation systems, demonstrating that no single metric class suffices for robust semantic equivalence assessment. Our empirical findings reveal critical blind spots in embedding-based approaches for numerical and negation perturbations, the inadequacy of exact matching metrics for linguistically diverse outputs, and fundamental trade-offs between scalability and semantic depth across traditional and LLM-based evaluators. These results underscore the necessity of hybrid evaluation frameworks that combine complementary metric classes, embedding-based similarity for general robustness, specialized fact-verification for numerical accuracy, and LLM-based judges for complex reasoning, while highlighting urgent open problems including combined pipeline evaluation, reasoning path traceability, and harmlessness assessment that demand attention in future RAG evaluation research.

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